1 Introduction

The edge of a wildfire is the most dangerous place for a firefighter. Legged robots can cross steep, rocky, smoky ground where vehicles stall and drones lose sight, yet a robot near a fire must know where the fire *will* be, not only where it is. We ask whether a learned spread forecast can keep a robot dog safe while it scouts, and whether the dog's own readings can improve that forecast.

2 Fire-spread forecast

We trained a gradient-boosted tree model (XGBoost[2]) on ∼246,000 VIIRS active-fire detections[3] from 30 major wildfires, using 27 features per 50 m cell: weather[4], terrain, land cover[5] and the local fire shape. The model scales the rate from a Rothermel spread equation[1], giving each cell's ignition probability every 15 min (Fig. 1).

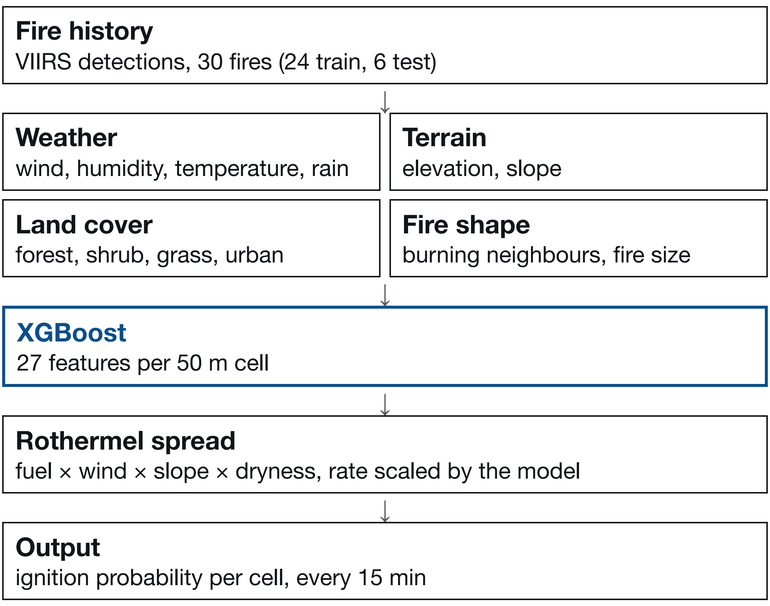


Fig. 1 | Forecast pipeline. Satellite fire history and conditions train the model; its output adjusts a physics-based spread rate.

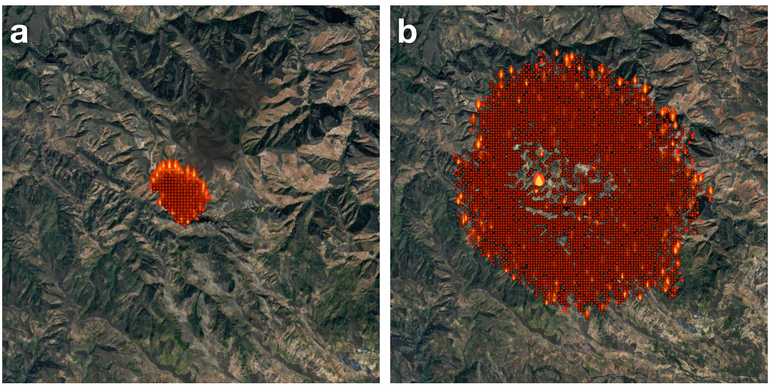


Fig. 2 | Browser simulator on real terrain. Simulated fire near Big Sur, Monterey County, after ∼4 h (**a**) and ∼1 day (**b**) of summer weather, rendered on Google Photorealistic 3D Tiles.

3 Forecast accuracy

On six held-out fires (USA, Bolivia, Siberia, Greece), 91% of cases are spread events, so we compare against a baseline that always predicts spread (Table 1). The model ranks at-risk cells well above chance and is better calibrated, but still over-predicts spread.

Table 1 | Accuracy on six unseen fires.

Metric	Model	Baseline
AUC-ROC (higher is better)	**0.75**	0.50
Brier score (lower is better)	**0.076**	0.083